	still holds true and the calculation in **(b)(iii)** and hence the ratio is the same.
6(a)	Faraday's law of electromagnetic induction states that the induced e.m.f. in a coil is directly proportional to the rate of change of magnetic flux linkage through the coil.
6(b)	When the wire moves, magnetic flux is cut. By Faraday's law, an e.m.f. is induced which is proportional to the rate of cutting of flux. By Lenz's law, due to the magnetic flux being cut in opposite directions as the wire oscillates up and down, if a current is to flow, it will flow in opposite directions and as such, the induced e.m.f. will alternate in directions.
6(c)	$E = Blv$ $= \left(250 \times 10^{-3}\right)(0.15)(0.80)$ $= 0.030\ V$
6(d)	0 V. The half of the wire from X to the folded point and the other half of the wire from Y to the folded point are both moving and cutting the magnetic flux at the same rate, thus the magnitude of the e.m.f. induced will be the same. In terms of polarity, X is at a potential lower than the folded end and Y is also at a potential lower than the folded end by the same magnitude. Thus, the net e.m.f. between ends X and Y is zero.
7(a)(i)	Photoelectric effect